SMOOTHNESS AT A PRIME OF A FINITE TYPE ALGEBRA IS FORMAL SMOOTHNESS OF THE LOCALISATION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The Artinian test criterion for smoothness of a finite type algebra at a prime. Algebra, Lemma \ref{algebra-lemma-smooth-test-artinian}, cited at proof lines 9667-9668. Gives the implication from formal smoothness to smoothness.
- A smooth algebra map is formally smooth. Algebra, Proposition \ref{algebra-proposition-smooth-formally-smooth}, cited at proof lines 9670-9671. Provides formal smoothness of the localised algebra.
- Formal smoothness is preserved by localisation, via the characterisation through the naive cotangent complex. Algebra, Proposition \ref{algebra-proposition-characterize-formally-smooth}, Algebra, Lemma \ref{algebra-lemma-NL-localize-bottom}, and Algebra, Lemma \ref{algebra-lemma-localize-NL}, cited at proof lines 9673-9680. Justifies passing to the localisations at the two primes.
- Formal smoothness upgrades to the adic topology. Lemma \ref{lemma-formally-smooth}, cited at proof lines 9681-9683. Turns algebraic formal smoothness into the adic statement of the lemma.

Source reference. Original lemma 07VH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10007–10007. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 02HT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3276–3276. Navigation pointer to a statement the

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Source reference. Original lemma 00TN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1081–1081. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 031J: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3986–3986. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07BS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9352–9352. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00S7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1029–1029. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07EC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9442–9442. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $A \to B$ be a finite type ring map with A Noetherian. Let $\mathfrak{q} \subset B$ be a prime ideal lying over $\mathfrak{p} \subset A$. The following are equivalent*

(1) *$A \to B$ is smooth at $\mathfrak{q}$, and*
(2) *$A_\mathfrak{p} \to B_\mathfrak{q}$ is formally smooth in the $\mathfrak{q}$-adic topology.*

Proof. The implication (2) $\Rightarrow$ (1) follows from Algebra, Lemma 02HT. Conversely, if $A \to B$ is smooth at $\mathfrak{q}$, then $A \to B_g$ is smooth for some $g \in B$, $g \notin \mathfrak{q}$. Then $A \to B_g$ is formally smooth by Algebra, Proposition 00TN. Hence $A_\mathfrak{p} \to B_\mathfrak{q}$ is formally smooth as localization preserves formal smoothness (for example by the criterion of Algebra, Proposition 031J and the fact that the cotangent complex behaves well with respect to localization, see Algebra, Lemmas 07BS and 00S7). Finally, Lemma 07EC implies that $A_\mathfrak{p} \to B_\mathfrak{q}$ is formally smooth in the $\mathfrak{q}$-adic topology. □